

Real-Time MAC Address Spoofing Detection using Wazuh SIEM with Automated Active Response and Email Alerts

CHAPTER 1

INTRODUCTION

1.1 Introduction

With the rapid growth of computer networks and internet-connected devices, cybersecurity has become an essential aspect of protecting organizational and personal data. Modern organizations rely on secure network infrastructures to ensure the confidentiality, integrity, and availability of information. As cyber threats continue to evolve, attackers use various techniques to bypass security mechanisms and gain unauthorized access to network resources. One such technique is **MAC (Media Access Control) address spoofing**, where an attacker changes the hardware address of a network interface to impersonate another device or evade network security controls.

A MAC address is a unique identifier assigned to a network interface card (NIC) by the manufacturer. It is commonly used for device identification, network access control, and communication within local area networks (LANs). However, using software tools, attackers can easily modify or spoof a MAC address. This enables them to bypass MAC-based filtering, impersonate trusted devices, avoid detection, and perform malicious activities while hiding their original identity.

Traditional network monitoring methods often fail to detect MAC address spoofing in real time. Many organizations rely on manual monitoring or basic logging mechanisms that provide limited visibility into such attacks. As a result, security administrators may not become aware of unauthorized MAC address changes until after a security incident has occurred.

To address this challenge, this project presents a **real-time MAC Address Spoofing Detection and Response System using Wazuh SIEM**. The system continuously monitors the MAC address of a network interface using a custom Python monitoring script. Whenever an unauthorized MAC address change is detected, the script generates a structured JSON event that is collected by the Wazuh agent and forwarded to the Wazuh Manager for analysis.

The Wazuh SIEM platform processes the collected event using custom detection rules and generates security alerts. The detected event is further mapped to the **MITRE ATT&CK Framework**, allowing security analysts to understand the adversarial technique involved. Additionally, the system executes an automated **Active Response**, records the investigation details, updates the Wazuh Dashboard, and sends an email notification to the administrator for immediate awareness.

This project demonstrates how open-source security technologies such as **Wazuh SIEM, Python, Linux, and Postfix** can be integrated to build an effective and automated security monitoring solution. The developed system improves incident detection, reduces response time, and provides centralized monitoring through an interactive dashboard and real-time alert notifications.

1.2 Objectives

The main objectives of this project are:

- To develop a real-time system for monitoring MAC address changes.
- To detect unauthorized MAC address spoofing attempts.
- To integrate the monitoring system with Wazuh SIEM.
- To generate custom security alerts for detected MAC changes.
- To map detected events to the MITRE ATT&CK framework.
- To perform automated Active Response after detection.
- To provide real-time email notifications to the administrator.
- To visualize security events using the Wazuh Dashboard.

1.3 Scope of the Project

This project focuses on detecting MAC address spoofing attacks within a Linux-based environment using Wazuh SIEM. It monitors changes in the MAC address of a network interface and generates alerts whenever unauthorized modifications are detected. The solution includes automated incident detection, dashboard visualization, email notifications, and Active Response to improve the efficiency of security monitoring. The project is suitable for educational environments, research laboratories, small organizations, and Security Operations Centers (SOCs) that require a low-cost, open-source monitoring solution.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing System

In many organizations, monitoring network devices is performed using traditional methods such as manual log inspection, basic network monitoring tools, or MAC address filtering. These methods are often insufficient to detect MAC address spoofing attacks in real time. Attackers can easily modify the MAC address of a network interface using readily available software tools, allowing them to bypass MAC-based authentication mechanisms and impersonate legitimate devices on the network.

Conventional monitoring systems generally record network events but do not automatically identify unauthorized MAC address changes or correlate them with potential security incidents. As a result, security administrators may not be immediately aware of malicious activities occurring within the network. Manual investigation also increases the response time, allowing attackers to continue their activities before appropriate action is taken.

Furthermore, many existing solutions do not provide centralized event management, automated response mechanisms, or real-time notification services. Without these capabilities, organizations face difficulties in quickly detecting, analyzing, and responding to MAC spoofing attacks.

Limitations of the Existing System

- No real-time MAC address monitoring.
- Manual detection of unauthorized MAC address changes.
- Delayed incident response.
- Lack of centralized security monitoring.
- No automated email notifications.
- Limited visualization of security events.
- Absence of automated investigation reports.
- No integration with the MITRE ATT&CK framework.

2.2 Proposed System

The proposed system provides an automated and centralized solution for detecting MAC address spoofing attacks using the Wazuh Security Information and Event Management (SIEM) platform. A custom Python script continuously monitors the MAC address of the network interface. Whenever a MAC address change is detected, the script collects relevant

system information such as the hostname, IP address, gateway, DNS server, and MAC address details, and stores the information in JSON format.

The generated log is monitored by the Wazuh agent and forwarded to the Wazuh Manager for analysis. Custom Wazuh detection rules identify the MAC address change and generate a security alert. The alert is mapped to the MITRE ATT&CK framework to provide additional threat intelligence and improve incident analysis.

After detection, an Active Response script is executed to record the incident details and generate an investigation report. Simultaneously, the Wazuh Dashboard is updated to display the alert, and an automated email notification is sent to the system administrator for immediate awareness and response.

The proposed system improves detection accuracy, reduces response time, and provides centralized monitoring of MAC spoofing incidents through an easy-to-use dashboard.

2.3 Advantages of the Proposed System

The proposed system offers several advantages over traditional monitoring approaches:

- Detects MAC address changes in real time.
- Provides automated security monitoring using Wazuh SIEM.
- Generates custom alerts for unauthorized MAC changes.
- Maps detected incidents to the MITRE ATT&CK framework.
- Executes automated Active Response for incident handling.
- Sends real-time email notifications to administrators.
- Displays alerts through the Wazuh Dashboard for centralized monitoring.
- Uses open-source technologies, reducing deployment cost.
- Can be easily extended to monitor multiple Linux systems.

2.4 Feasibility Study

Technical Feasibility

The project is technically feasible as it is implemented using open-source technologies including Python, Wazuh SIEM, Ubuntu Server, Kali Linux, and Postfix. These technologies are freely available and compatible with standard computing environments.

Operational Feasibility

The system is easy to operate and requires minimal user intervention. Once configured, it continuously monitors MAC address changes, automatically detects suspicious activities, and notifies the administrator without manual monitoring.

Economic Feasibility

The proposed system is cost-effective because it relies entirely on open-source software. No commercial licenses are required, making it suitable for educational institutions, research laboratories, and small organizations with limited budgets.

CHAPTER 3

SYSTEM DESIGN

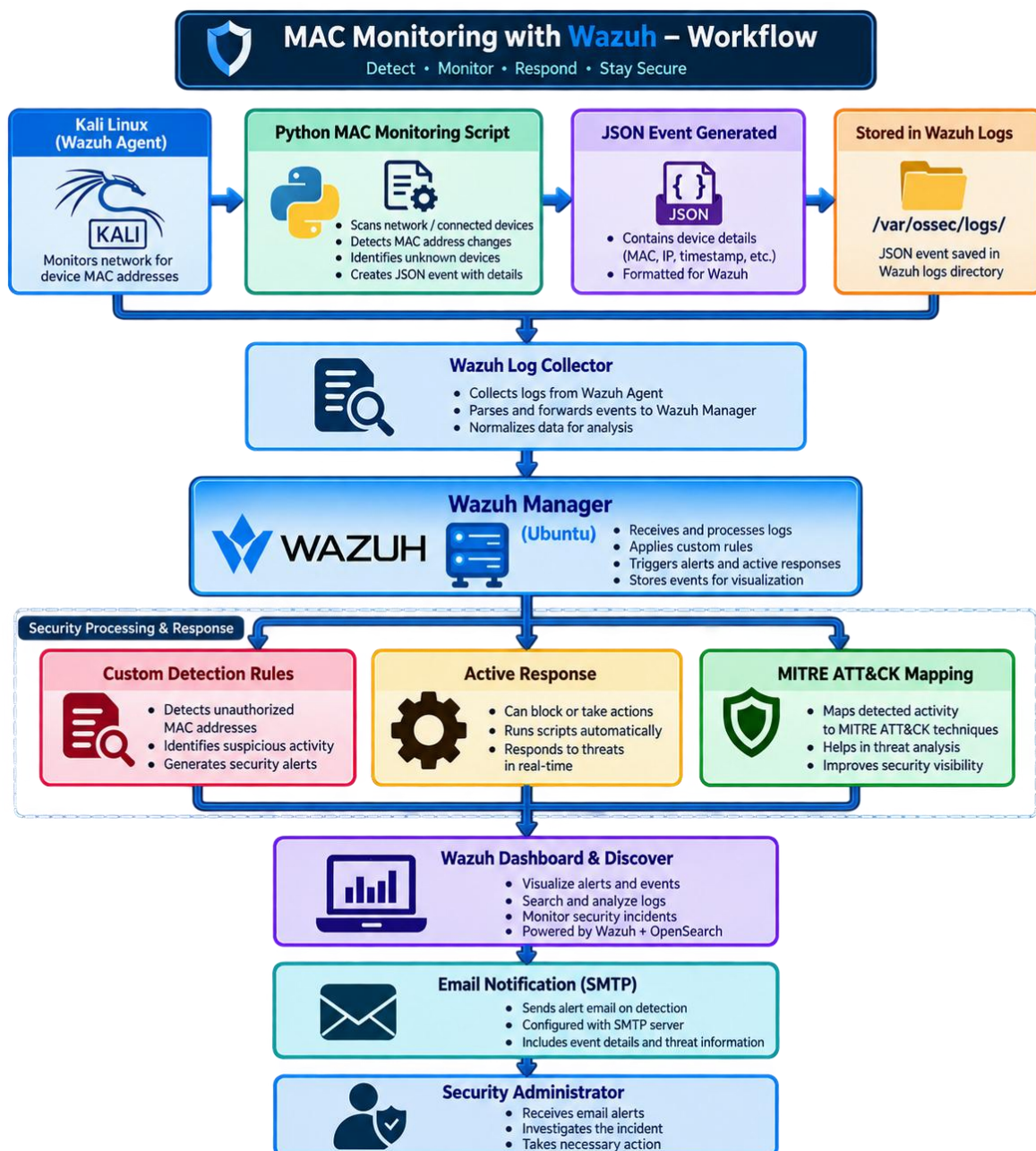
3.1 System Overview

The proposed system is designed to detect unauthorized MAC address changes in real time using the Wazuh Security Information and Event Management (SIEM) platform. The system continuously monitors the MAC address of a Linux machine using a custom Python script. When a MAC address change is detected, the script collects network information such as the current MAC address, previous MAC address, hostname, IP address, gateway, DNS server, and timestamp.

The collected information is stored in JSON format and written to the Wazuh log file. The Wazuh agent forwards the log to the Wazuh Manager, where custom detection rules analyze the event and generate a security alert. The detected event is mapped to the MITRE ATT&CK framework and displayed on the Wazuh Dashboard. Simultaneously, an Active Response script is executed to record the incident, and an email notification is sent to the administrator.

This automated workflow enables rapid detection, analysis, and response to MAC spoofing attacks without requiring continuous manual monitoring.

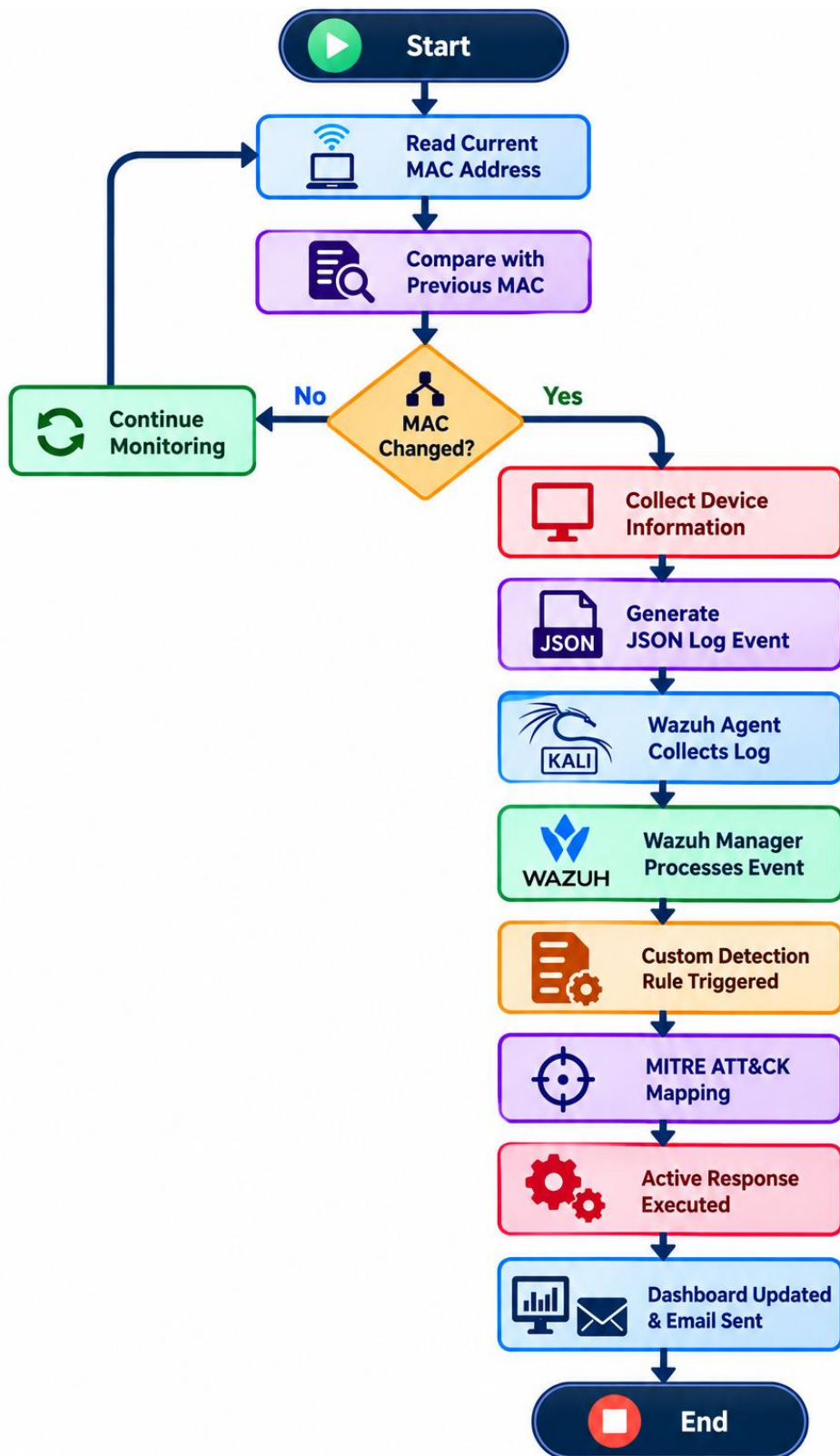
3.2 System Architecture



Explanation

The Kali Linux machine acts as the monitored endpoint and runs both the Wazuh Agent and the custom Python monitoring script. Whenever a MAC address change is detected, the monitoring script generates a JSON log containing detailed system information. The Wazuh Agent forwards this log to the Wazuh Manager running on Ubuntu Server. Custom Wazuh rules analyze the event, trigger alerts, perform MITRE ATT&CK mapping, execute the Active Response script, and update the Wazuh Dashboard. Finally, an email notification is sent to the administrator for immediate action.

3.3 Workflow of the Proposed System



Explanation

The monitoring process begins by reading the current MAC address of the network interface. The current value is compared with the previously stored MAC address. If no changes are detected, the monitoring process continues. If a change is detected, the system collects device and network information, generates a JSON event, and forwards it to the Wazuh Manager through the Wazuh Agent. Wazuh applies custom detection rules, maps the incident to the MITRE ATT&CK framework, executes the Active Response script, updates the dashboard, and sends an email notification to the administrator.

3.4 Modules of the System

The proposed system consists of the following modules:

Module 1 – MAC Address Monitoring

The Python script continuously monitors the network interface and detects changes in the MAC address.

Module 2 – Log Generation

After detecting a MAC address change, the system collects network information and stores it as a structured JSON event.

Module 3 – Wazuh SIEM Integration

The Wazuh Agent forwards the generated log to the Wazuh Manager for analysis and event correlation.

Module 4 – Alert Generation

Custom Wazuh rules identify MAC address changes and generate security alerts with the configured severity level.

Module 5 – MITRE ATT&CK Mapping

The detected event is mapped to the MITRE ATT&CK framework to provide additional context about the attack technique.

Module 6 – Active Response

An automated response script records the incident details and generates an investigation report for future analysis.

Module 7 – Dashboard and Email Notification

The Wazuh Dashboard displays the detected alerts, while an email notification is automatically sent to the administrator whenever a MAC spoofing event is detected.

Chapter 4

Implementation

4.1 System Setup

The project was implemented using a Wazuh SIEM server running on Ubuntu and a Kali Linux machine as the monitored endpoint. The Wazuh Agent was installed on Kali Linux to forward security events to the Wazuh Manager. The Wazuh Dashboard was used to visualize alerts and monitor the system.

Tools Used

- Ubuntu 24.04 – Wazuh Server
- Kali Linux – Monitored Endpoint
- Wazuh SIEM 4.14.7
- Python 3
- Bash Shell Script
- macchanger
- Postfix (Email Notification)
- OpenSearch Dashboard

4.2 Project Workflow

The project continuously monitors the MAC address of the network interface.

Workflow:

1. Read the current MAC address.
2. Compare it with the previous MAC address.
3. Detect any MAC address change.
4. Generate a JSON investigation report.
5. Send logs to Wazuh Agent.
6. Wazuh Manager analyzes the event.
7. Trigger custom detection rule.
8. Generate alert on the Dashboard.
9. Send email notification.
10. Execute Active Response (optional).

4.3 Python Monitoring Script

A custom Python script (`mac_monitor.py`) was developed to monitor the MAC address continuously. Whenever a change is detected, it collects additional forensic information such as:

- Old MAC Address
- New MAC Address
- Permanent MAC
- IP Address
- Default Gateway
- DNS Server
- Hostname
- Timestamp
- Vendor Information
- Randomized Status
- Change Count

The information is stored in JSON format and forwarded to Wazuh for analysis.

4.4 Wazuh Log Collection

The generated JSON logs are stored in:

`/var/ossec/logs/mac_monitor.log`

The Wazuh Agent monitors this log file and forwards every new event to the Wazuh Manager for correlation and analysis.

4.5 Custom Detection Rule

A custom Wazuh rule was created to detect MAC address modification events.

The rule:

- Detects `MAC_INVESTIGATION`
- Assigns Rule ID **100510**
- Generates High Severity Alerts
- Maps the event to MITRE ATT&CK

This allows the security team to quickly identify unauthorized MAC spoofing attempts.

4.6 Dashboard Visualization

Custom visualizations were created in the Wazuh Dashboard to monitor:

- MAC Address Change Alerts
- Events by Agent
- Timeline of MAC Changes
- Recent Investigation Events

These visualizations provide real-time monitoring and improve incident visibility.

4.7 Email Notification

Postfix SMTP was configured to send email alerts whenever a MAC address change is detected.

The notification includes:

- Timestamp
- Hostname
- Interface
- Old MAC
- New MAC
- Alert Level

This ensures administrators are immediately informed of suspicious activity.

4.8 Active Response

When a MAC spoofing event is detected, Wazuh executes an Active Response script that logs the incident and can optionally restore the original MAC address or perform other response actions.

The response helps reduce the impact of unauthorized MAC address changes.

4.9 Testing and Validation

The project was tested by intentionally changing the MAC address using the macchanger utility.

Test Steps:

1. Run the monitoring script.
2. Change the MAC address.
3. Verify JSON log generation.
4. Confirm alert appears in Wazuh Dashboard.
5. Verify email notification is received.
6. Check Active Response execution.

The results confirmed that the system successfully detected, logged, alerted, and responded to MAC spoofing events.

Chapter 5

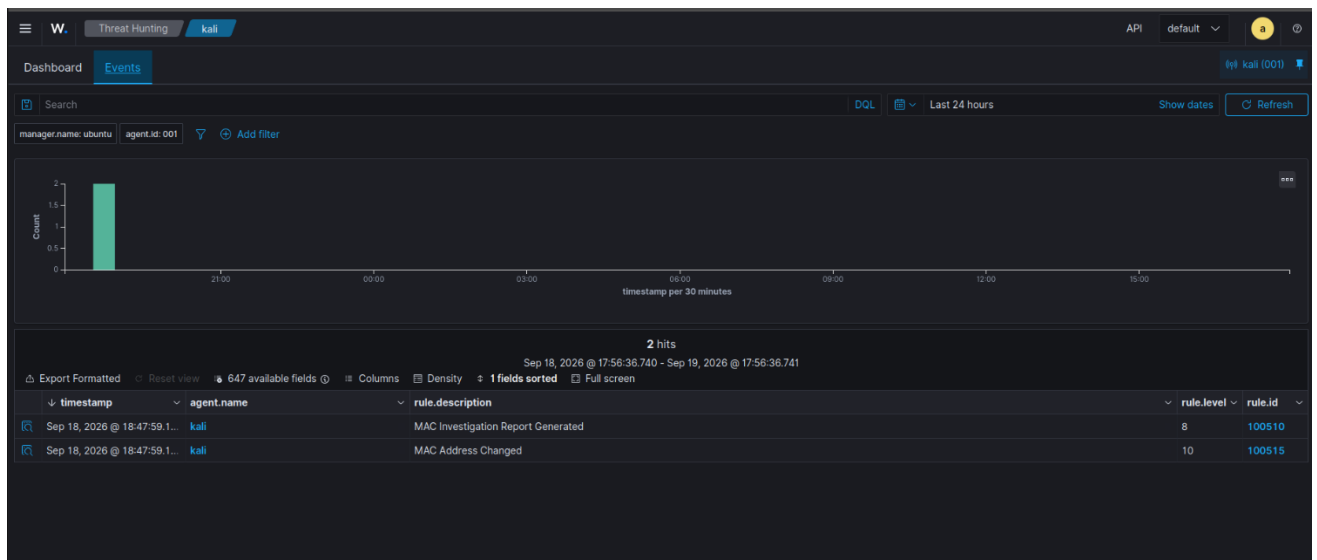
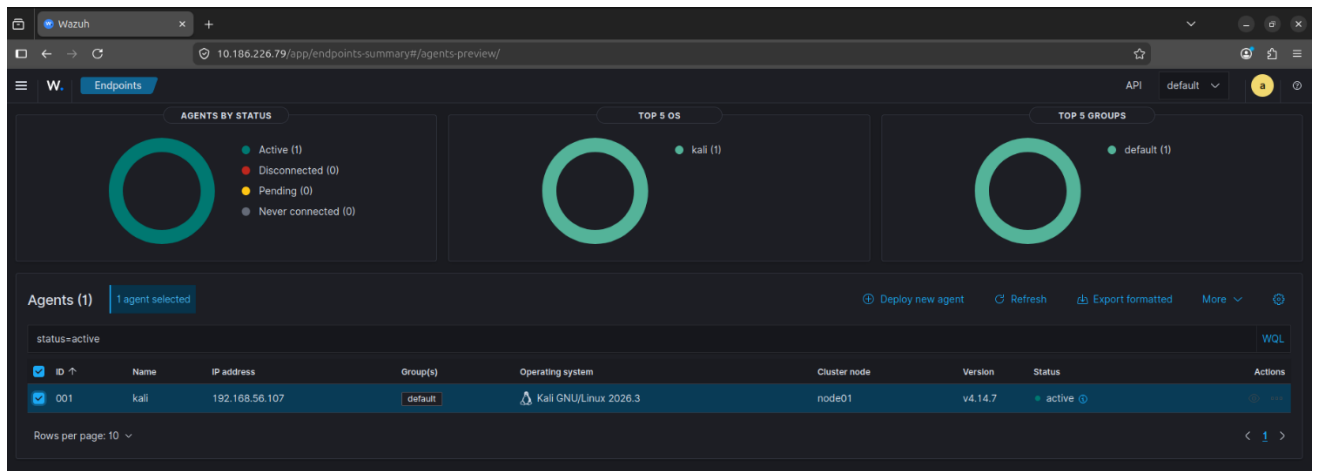
Results & Testing

Screenshots

```
(root@kali)-[/home/kali]
# sudo ip link set eth0 down
sudo macchanger -r eth0
sudo ip link set eth0 up
```

```
(root@kali)-[/home/kali]
# sudo ip link set eth0 down
sudo macchanger -r eth0
sudo ip link set eth0 up
Current MAC: 08:00:27:8a:35:d2 (CADMUS COMPUTER SYSTEMS)
Permanent MAC: 08:00:27:8a:35:d2 (CADMUS COMPUTER SYSTEMS)
New MAC: 7e:b6:9f:15:d1:74 (unknown)
```

```
(root@kali)-[/var/ossec/etc]
# sudo tail -20 /var/ossec/logs/mac_monitor.log
{"event": "MAC_INVESTIGATION", "timestamp": "2026-09-16T20:19:41.520214", "hostname": "kali", "interface": "eth0", "ip_address": "10.186.226.6", "default_gat
:2d", "permanent_mac": "3a:bc:68:77:14:10", "vendor": "3a:bc:68:77:14:10", "randomized": "YES", "change_count": 1}
{"event": "MAC_INVESTIGATION", "timestamp": "2026-09-16T20:27:58.337279", "hostname": "kali", "interface": "eth0", "ip_address": "10.186.226.6", "default_gat
permanent_mac": "72:8a:6b:09:69:2d", "vendor": "3a:bc:68:77:14:10", "randomized": "YES", "change_count": 2}
{"event": "MAC_INVESTIGATION", "timestamp": "2026-09-16T20:28:03.343749", "hostname": "kali", "interface": "eth0", "ip_address": "10.186.226.227", "default_g
1b:b5", "permanent_mac": "42:a5:bf:2c:73:a9", "vendor": "3a:bc:68:77:14:10", "randomized": "YES", "change_count": 3}
{"event": "MAC_INVESTIGATION", "timestamp": "2026-09-16T20:28:23.395393", "hostname": "kali", "interface": "eth0", "ip_address": "10.186.226.227", "default_g
d6:01", "permanent_mac": "b6:70:db:12:1b:b5", "vendor": "3a:bc:68:77:14:10", "randomized": "YES", "change_count": 4}
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9c:78", "permanent_mac": "42:03:7f:db:d6:01", "vendor": "3a:bc:68:77:14:10", "randomized": "YES", "change_count": 5}
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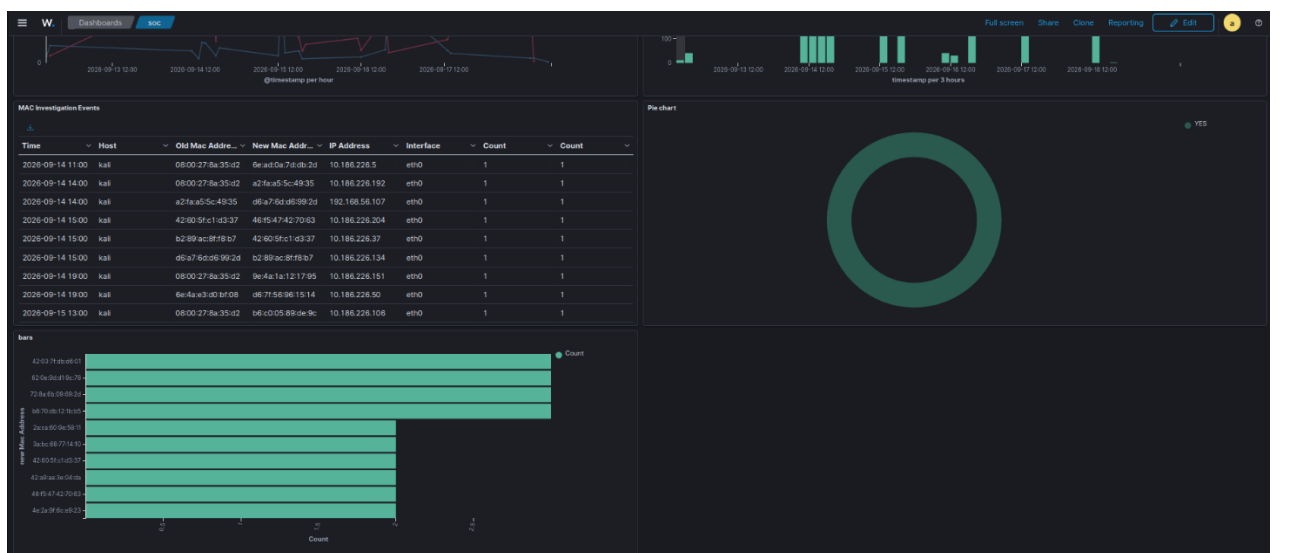
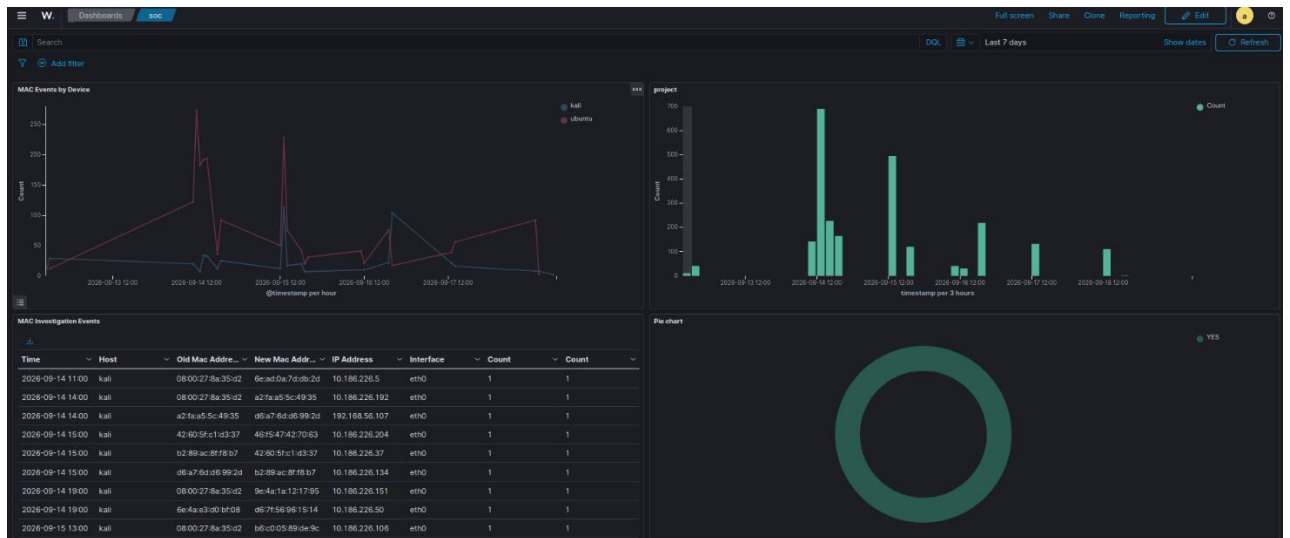


Dashboards

Search...

Title	Type	Description	Last updated	Actions
soc	Dashboard		Sep 18, 2026 @ 13:43:17.770	

Rows per page: 20



soc_analyst  

An offer awaits 

Find a conversation...

Huddles

Channels

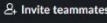
- # all-socanalyst**
- # new-channel
- # security-alerts
- # social
- + Add channels

Direct messages

- Barath purushothaman you
- + Invite people

Agents & apps

- sample
- Slack
- Google Calendar
- Google Drive

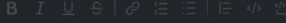
all-socanalyst     

Messages  

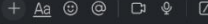
User: root
Old MAC: e6:cf:1c:e6:30:c6
New MAC: f6:ac:df:4e:f5:a5
IP: 10.186.226.173

2:51  **MAC Address Changed**
Host: kali
User: root
Old MAC: f6:ac:df:4e:f5:a5
New MAC: 52:4a:bc:21:94:20
IP: 10.186.226.105

 **Incoming-webhook** APP 6:48 PM
 **MAC Address Changed**
Host: kali
User: root
Old MAC: 08:00:27:8a:35:d2
New MAC: 7e:b6:9f:15:d1:74
IP: 10.186.226.45



Message #all-socanalyst



Wazuh notification - (kali) any - Alert level 10



Wazuh <barathpurushothaman28@gmail.com>
to me ▾

Fri 18 Sept, 18:48 (23 hours ago)    

Wazuh Notification.
2026 Sep 18 13:17:59

Received From: (kali) any->/var/log/mac_monitor.log
Rule: 100515 fired (level 10) -> "MAC Address Changed"
Portion of the log(s):

```
HOST=kali USER=root INTERFACE=eth0 PERMANENT_MAC=08:00:27:8a:35:d2 OLD_MAC=08:00:27:8a:35:d2 NEW_MAC=7e:b6:9f:15:d1:74 VENDOR=CADMUS_COMPUTER_SYSTEMS
RANDOMIZED=YES CHANGE_COUNT=1 EVENT=MAC_ADDRESS_CHANGED
hostname: kali
mac_user: root
interface: eth0
permanent_mac: 08:00:27:8a:35:d2
old_mac: 08:00:27:8a:35:d2
new_mac: 7e:b6:9f:15:d1:74
vendor: CADMUS_COMPUTER_SYSTEMS
randomized: YES
change_count: 1
event: MAC_ADDRESS_CHANGED
```

Chapter 6

Conclusion

6.1 Conclusion

This project successfully developed a **MAC Address Change Detection and Investigation System** using **Wazuh SIEM**. The system continuously monitors the MAC address of a Linux endpoint and detects unauthorized MAC address changes in real time.

When a MAC address change is detected, the system collects investigation details such as the old MAC address, new MAC address, IP address, gateway, DNS server, hostname, and timestamp. These details are forwarded to Wazuh, where custom detection rules generate security alerts. The alerts are displayed on the Wazuh Dashboard, email notifications are sent to the administrator, and Active Response scripts can be executed automatically.

The project demonstrates how SIEM technology can be used to improve endpoint monitoring and detect MAC spoofing attacks quickly. It also provides a practical solution for enhancing network security through real-time detection, investigation, visualization, and automated response.

6.2 Future Enhancements

The following features can be added in future versions:

- Detect spoofing across multiple network interfaces.
- Store historical MAC address information in a database.
- Integrate threat intelligence feeds for vendor verification.
- Develop a web-based investigation dashboard.
- Support Windows and macOS endpoints.
- Integrate with Slack or Microsoft Teams for instant notifications.
- Apply machine learning to detect abnormal MAC address behavior.
- Generate automated forensic investigation reports.

References

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2. MITRE ATT&CK Framework. <https://attack.mitre.org/>
3. Kali Linux Documentation. <https://www.kali.org/docs/>
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